

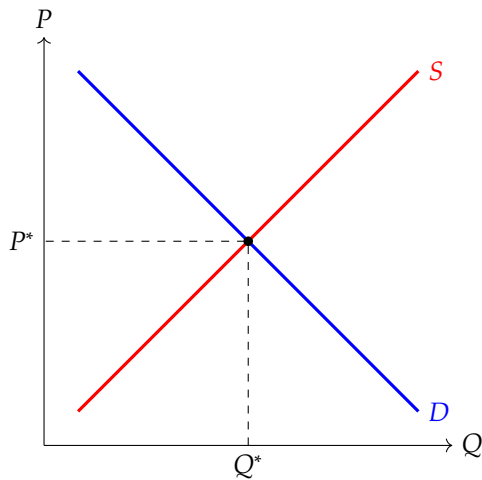
TikZ：経済学でよく使う図

山田 知明

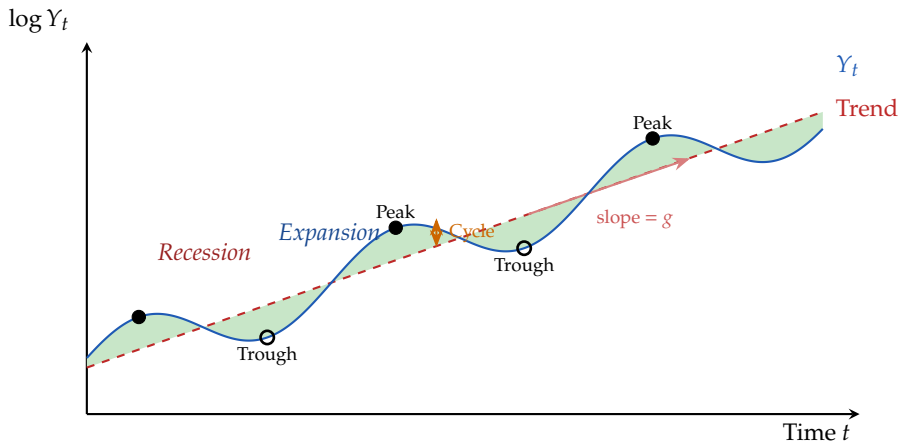
明治大学 商学部

2026 年 4 月 7 日

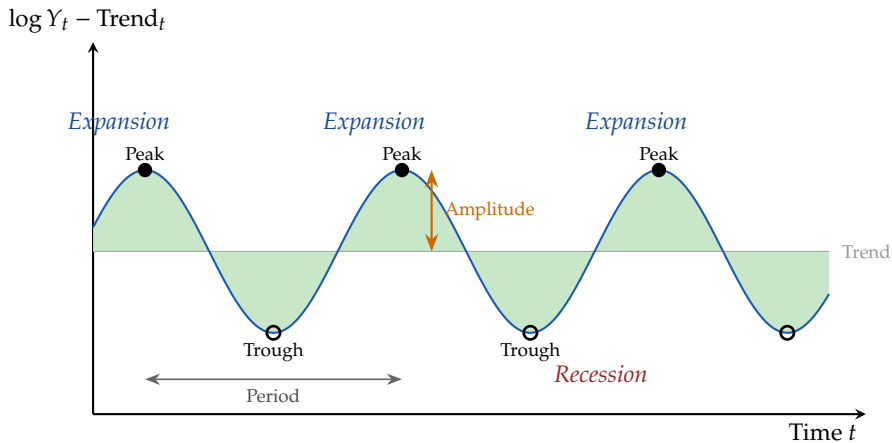
需要と供給



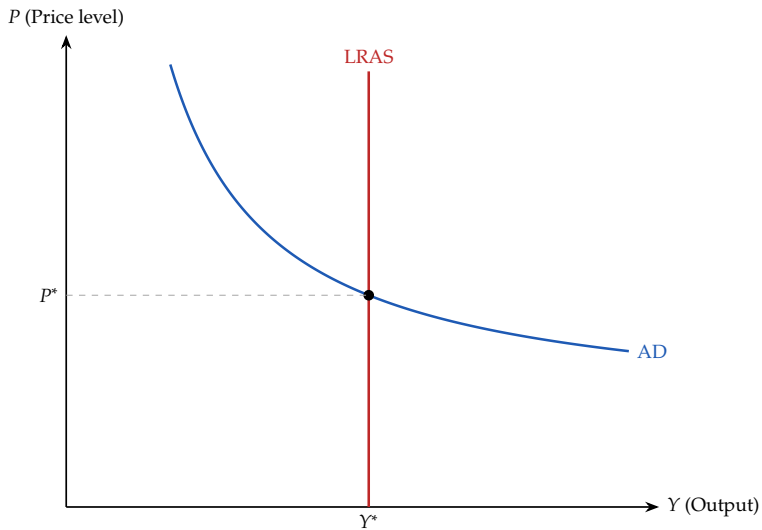
経済成長と景気循環



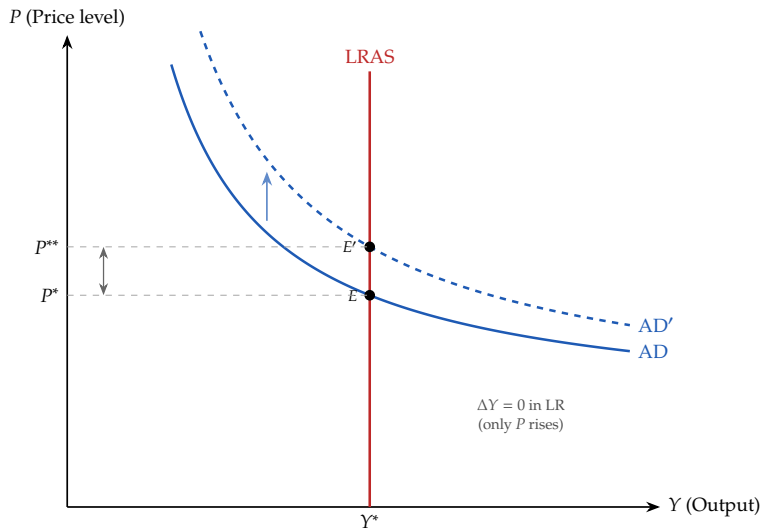
景気循環



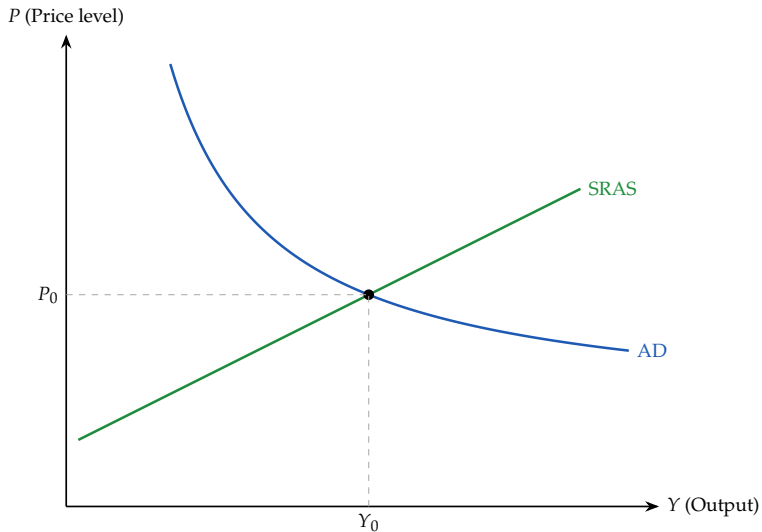
総需要・総供給分析：長期均衡



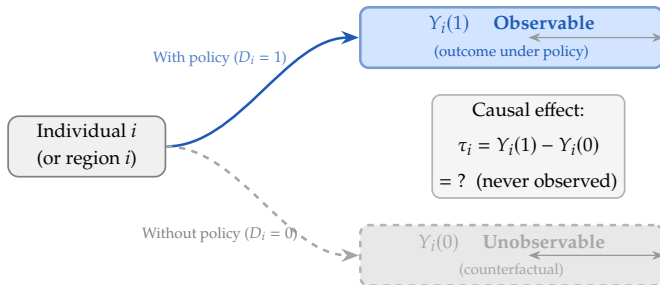
総需要・総供給分析：長期均衡



総需要・総供給分析：短期均衡

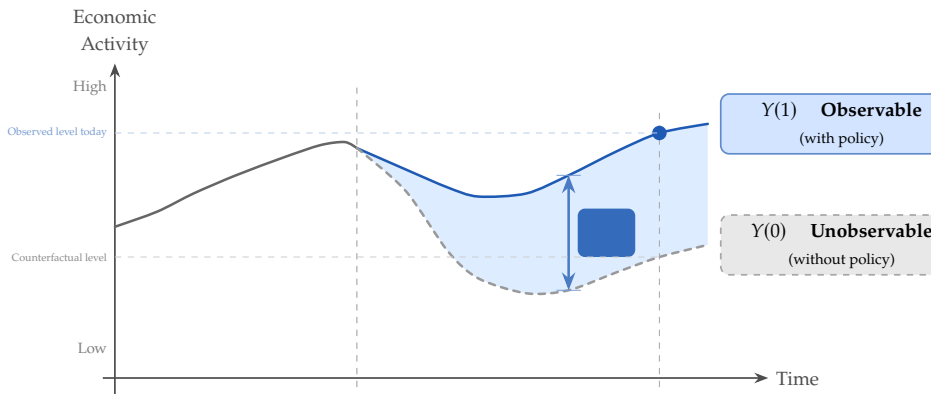


政策評価がなぜ難しいのか？：因果推論の根本問題



Only one state is ever observed.
This is the *Fundamental Problem of Causal Inference*.

政策評価がなぜ難しいのか？：景気対策の評価

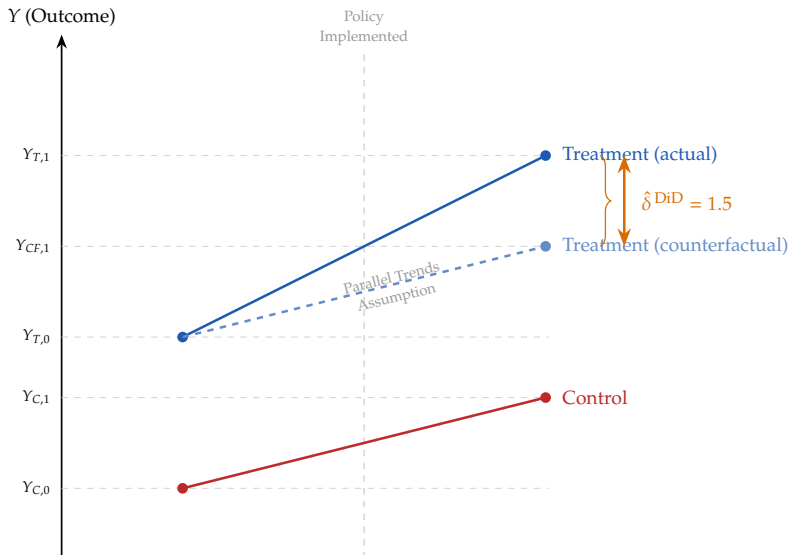


The economy looks “not so bad” only because the policy worked.

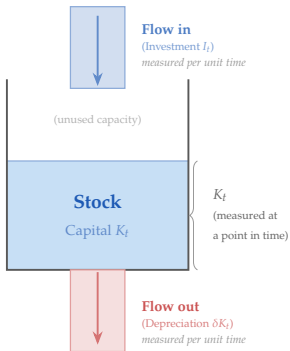
Without intervention, the downturn would have been far deeper.

This gap — $Y(1) - Y(0)$ — is the *causal effect*, but $Y(0)$ is never observed.

Difference-in-Differences 分析のイメージ



ストックとフローの関係



Examples in Economics

Stock	Flow
Capital K	Investment I , Depreciation δK
Population N	Births, Deaths
Wealth W	Saving, Consumption
Government debt	Budget deficit / surplus
Water reservoir	Rainfall, Outflow

$$\underbrace{K_{t+1}}_{\text{Stock (next period)}} = \underbrace{K_t}_{\text{Stock}} + \underbrace{I_t}_{\text{Flow in}} - \underbrace{\delta K_t}_{\text{Flow out}}$$

三面等価の原則

Three Equivalent Measures of GDP (三面等価)

